# Early ADR Signal Detection With Transformer-Based Deep Reinforcement Learning

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***Abstract*—Early detection of adverse drug reactions (ADRs) is critical for post-market drug safety, yet remains challenging due to the noisy, self-reported nature of pharmacovigilance data, where labels are both sparse and potentially unknown. Using quarterly records from the Federal Adverse Event Reporting System (FAERS), we propose a transformer-based reinforcement learning framework for early ADR signal detection. The problem is formulated as a Partially Observable Markov Decision Process (POMDP), where a transformer-encoder architecture is used to learn the latent state belief given an observation history. A novel time-aware reward function is introduced to enable the agent to learn temporal decision making policies which encourage early detection without sacrificing precision and recall. To address the absence of ground-truth detection times, we construct surrogate temporal labels using an ensemble of classical signal detection methods widely adopted by many agencies. Experimental results on FAERS demonstrate that the proposed approach achieves 94% early detection with respect to 70% correctly detected positives, while still maintaining competitive precision.**

***Index Terms*—Adverse Drug Reactions, FAERS, Pharmacovigilance, Reinforcement Learning, Transformer, Signal Detection**

## I. Introduction

Adverse Drug Reactions (ADRs) describe harmful side effects associated with various prescribed drugs or treatments. While ADRs are ideally identified during clinical trials, many drug–reaction pairs go undetected and must be identified post-market as quickly as possible. For this reason, the Federal Adverse Events Reporting System (FAERS) allows individuals to report cases of reactions that may be associated with the prescription of a given drug.

ADR signals are commonly represented using contingency tables summarizing co-occurrence statistics between drugs and reactions. Given quarterly report records from FAERS, many statistical signal detection methods are commonly used to identify significant drug-reaction associations. Due to the inherent difficulty of working with noisy and sparsely labeled pharmacovigilance data, the goal is the flag potential ADRs for human review, limiting resources wasted on false positives and accounting for the consequences of missing a positive reaction association. Recently, deep learning methods have been used to improve the quality and generalization of learned characteristics of ADR signals as they emerge over many quarters of reporting.

The goal of this paper is to design a model which can detect a statistically associated ADR signal as early as possible from the onset of a new drug–reaction pair. In application, this allows ADRs to be detected without needing extensive observation history, allowing human reviewers to be sent potentially associated pairs earlier. Since we only have positive drug-reaction labels as of present, this requires a reasonable estimation of the quarter a method would likely have been detected with traditional methods. A model-free reinforcement learning agent will be trained by implementing a transformer to learn latent signal dynamics, supported by a reward structure encouraging the agent to balance early detection with temporal decision-making confidence.

## II. Related Works

Temporal modeling and decision making is a common paradigm in reinforcement learning. Temporal logic (TL) is utilized by Li et al. [1] to define rewards which may depend on the trajectory or timestep itself, not exclusively a given state and action. Non-stationary environments or evolving transitions can be solved with sliding windows that approximate stationarity by algorithmically discarding long–range prior observations [2] [3]. Most notably, Chen et al. [4] demonstrate that even robust sliding window algorithms can be well approximated by deep learning architectures, namely transformers and recurrent networks.

Another line of work addresses decision making under partial observability, where the agent must infer a latent state from an observation history. Recurrent networks and transformers again demonstrate robust approximation for belief states or update rules, even without

explicitly encoding a statistical observation model or complex POMDP algorithms [5] [6] [7].

Most closely related to our implementation is the ADRP–DQPU proposed by Chung & Lin [8]. ADR detection is formulated in a reinforcement learning environment, where a double Q-learning network is used to classify a drug–reaction pair as a positive ADR association or leave unlabeled based on reports from a single quarter. This work improves on standard classification methods employed over FAERS, which generally use statistical detectors, gradient boosting methods, or logistic regression [9] [10] [11].

## III. Datasets

### A. FAERS

FAERS [12] is a publicly available repository of adverse event reports submitted to the U.S. Food and Drug Administration. While FAERS provides large-scale, real-world reports, it is inherently noisy due to its reliance on spontaneous and self-reported submissions. Care is needed to remove duplicate reports and ensure drugs are normalized and can be aggregated accurately at the ingredient level.

Additionally, FAERS does not include any labels for known ADR pairs. The dataset strictly covers the self-reported entries submitted to the FDA. Therefore, three additional datasets are required and described in the following subsections.

### B. RxNorm & UMLS

With the goal of mapping reported drug strings to a medically uniform identifier at the ingredient level, we utilize the RxNorm [13] repository to encode normalized strings to RxNorm Concept Identifiers (RXCUI). Although this provides a standardized identifier for the various active ingredients in FAERS, our labeling tables will require drugs encoded with their PubChem [14] CID. RxNorm concepts are further mapped to chemical compound identifiers with FDA Unique Ingredient Identifiers (UNII) contained in the `RXNCONSO.RRF` file by selecting entries with source `MTHSPL`. Finally, UNII identifiers are mapped to PubChem compound identifiers (CID) using the PubChem synonym table (`CID-Synonym-filtered.gz`). This enables conversion from clinically normalized RxNorm concepts to chemical-level identifiers, allowing consistent integration with SIDER and BioSNAP, which will be used for labeling.

### C. SIDER & BioSNAP

In order to mediate the issue of unknown labels for the FAERS dataset, two additional sources are used. SIDER [15] and BioSNAP [16] include known relationships between associated drug–reaction pairs. These pairs use the MedDRA PT for reactions and PubChem CID for drug names. Using our derived mappings, drug–reaction pairs are labeled as positive if they appear in SIDER or BioSNAP, and otherwise treated as unlabeled. It is important to emphasize that unlabeled pairs do not imply negative associations, but rather reflect the incomplete nature of existing pharmacovigilance databases. This pipeline is a robust implementation of typical data engineering practices used to prepare drug relationship data for training machine learning models [17] [18].

### D. True Detection Time Estimation

While we do not know precisely which quarter an ADR association was first established, we may estimate a label $\tau^\star$ based on standard signal detection methods. The proposed methods are Reporting Odds Ratio (ROR), Proportional Reporting Ratio (PRR), Bayesian Confidence Propagation Neural Network (BCPNN), Sequential Probability Ratio Test (SPRT), Yule's Q, and the Medicines and Healthcare products Regulatory Agency (MHRA) method. This ensemble selection is motivated by [8]. These statistical methods are standard in detecting ADRs on FAERS. While our estimated $\tau^\star$ label does not reflect a ground-truth detection time, it stands as a reasonable surrogate label for a method which will not rely on the exact accuracy of $\tau^\star$, but instead learns temporal decision-making policies that balance early detection with predictive reliability.

Let $M$ denote the number of signal detection methods, and index each method by $m \in \{1, \dots, M\}$. For each method $m$ and quarter $t$, define the indicator function

$$q_m(t) = \begin{cases} 1, & \text{method } m \text{ detects signal} \\ 0, & \text{otherwise.} \end{cases}$$

We then define

$$k_t = \sum_{m=1}^{M} q_m(t),$$

which represents the total number of methods that detect a signal at time $t$.

We say a drug–reaction pair exhibits an ensemble ADR signal at time $t$ if

$$k_t \geq \kappa,$$

where $\kappa$ is a predefined threshold. We now define the variables:

$$t_n = \min \{\, t \;:\; k_j \geq \kappa \;\; \forall\, j = t, \dots, t + n - 1 \,\}$$

which provide a reasonable enforcement of sustained, rather than one-off detection.

Finally:

$$\tau^\star = \text{median}(t_2, t_3, t_4) \tag{1}$$

If fewer than two of $t_2, t_3, t_4$ exist, $\tau^\star$ is set to be the most recent quarter. The median provides a robust consensus estimator and is less sensitive to noise. Labels for $\tau^\star$ are only augmented over positive labels, as unlabeled pairs inherently have no detection time.

## IV. Methodology

### A. Partially Observable Markov Decision Process Formulation

The problem is formulated as a partially observable Markov decision process (POMDP) $(\mathcal{S}, \mathcal{A}, \mathcal{O}, \mathcal{P}, \mathcal{R}, \mathcal{Z})$. In this setting, the true underlying ADR signal dynamics are not directly observed, and the agent must act based on noisy observations derived from FAERS reports.

*1) State:* We introduce a state space $\mathcal{S}$ to represent the underlying signal dynamics which are not directly observable. For a given drug–reaction pair $(D, R)$, the latent state at quarter $t$ is denoted by

$$s_t = (D, R, y, \eta_t) \in \mathcal{S},$$

where $y \in \{0, 1\}$ is a binary variable indicating whether a true ADR association exists for the pair, and $\eta_t \in \mathbb{R}^k$ is a latent feature vector which, though not observable, captures underlying signal properties. It is our goal to learn the feature vector $\eta_t$ and subsequently the label $y$.

*2) Observation:* A given drug–reaction pair includes quarterly records enumerated by $t = 1, \ldots, T$. For each chronological timestep in the episode, the agent observes $o_t \in \mathcal{O}$ defined as

$$o_t = (D, R, a_t, b_t, c_t, d_t),$$

where $D$ and $R$ denote a drug–reaction pair, and $a_t, b_t, c_t, d_t$ correspond to the entries of a standard $2 \times 2$ contingency table of aggregated data up to quarter $t$ constructed from FAERS reports. The contingency table is shown in tabular form below:

TABLE I
Contingency table for drug $D$ and reaction $R$ up to quarter $t$.

| | Reaction $R$ | Not $R$ |
|---|---|---|
| Drug $D$ | $a_t$ | $b_t$ |
| Not $D$ | $c_t$ | $d_t$ |

The table corresponds to a particular and unique drug–reaction pair. Specifically, $a$ denotes the number of reports in which drug $D$ is administered and reaction $R$ occurs, $b$ denotes the number of reports in which drug $D$ is administered and some reaction other than $R$ occurs, $c$ denotes the number of reports in which a drug other than $D$ is administered and reaction $R$ occurs, and $d$ denotes the number of reports in which neither drug $D$ nor reaction $R$ co-occur. Reports are counted by their ID in the FAERS database, therefore the table is constructed at the unique report level.

*3) Transition Model:* The state evolves according to a transition model

$$P(s_{t+1} \mid s_t, a_t).$$

The agent's actions do not influence the underlying ADR signal dynamics, but only determine whether the episode continues. That is, the signal evolves naturally as medical reports come in, irrespective of choices the agent makes. Therefore:

$$P(s_{t+1} \mid s_t) = P(\eta_{t+1} \mid \eta_t, y),$$

*4) Observation Model:* We denote that observations are generated from the true state according to an observation model

$$\mathcal{Z}(o_t \mid s_t) = P(o_t \mid s_t),$$

which represents the noise and reporting bias present in FAERS records relative to true signal dynamics. Therefore,

$$\mathcal{Z}(o_t \mid s_t) = P(a_t, b_t, c_t, d_t \mid D, R, y, \eta_t).$$

While the true model is unknown to us, an implicit state belief can still be approximated from the observation history, as we will detail in later sections.

*5) Action Space:* The enumerated action space consists of three possible actions for the agent, $\mathcal{A} = \{0, 1, 2\}$. Action 0 corresponds to deferring a decision in lieu of more information from following timesteps. Action 1 allows the agent to flag the pair $(D, R)$ as an ADR at quarter $t$, and Action 2 allows the agent to exit the episode early, asserting that no relevant ADR statistics are present.

*6) Reward:* At each timestep $t$, the agent selects an action $a_t \in \{0, 1, 2\}$ corresponding to *defer*, *flag ADR*, and *terminate episode*, respectively. The reward is defined as:

$$r_t = \begin{cases} \text{Wait:} & \rho\, e^{-t/T_1} - \lambda \\ \text{TP:} & w_{\text{pos}} \cdot \exp\!\left(-\frac{(t-\alpha\tau^\star)^2}{2\sigma^2}\right) \\ \text{FP:} & -\ w_{\text{neg}} \left(1 + \beta_1 e^{-t/T_2}\right) \\ \text{TN:} & w_{\text{neg}} \cdot \left(1 - \beta_2 e^{-t/T_3}\right) \\ \text{FN:} & -\ w_{\text{pos}} \left(2 - \beta_3 e^{-t/\tau^\star}\right) \end{cases}$$

where TP, FP, TN, and FN denote a true positive, false positve, true negative, and false negative, respectively. The weights $w_{pos}$ and $w_{neg}$ are class weights which account for the data imbalance and are derived from the observed ratio of positives to negatives in the dataset.

It should also be restated that $\tau^\star$ is a label which exists only for positives samples, therefore, TP and FN rewards may depend on $\tau^\star$.

The reward function is designed to encourage early and accurate ADR detection while accounting for class imbalance and temporal uncertainty. A decaying wait reward promotes exploration early in the sequence but discourages indefinite deferral. True positives are shaped by a Gaussian centered early at $\alpha\tau^\star, \alpha \in (0,1)$, rewarding detections within an early window of true detection time with a smooth and differentiable function. False positives incur a time-dependent penalty that is strongest early and attenuates over time, discouraging premature flagging. True negatives receive an increasing reward as more evidence accumulates, reflecting growing confidence in the absence of a signal and promoting stronger recall. False negatives are penalized with both a constant and time-dependent component, ensuring that missed detections are always discouraged and increasingly so when occurring after $\tau^\star$. While the reward is shown with general hyperparameters for an exponential growth or decay function, our selections are detailed in B.

### B. Double Q-Learning

To learn stable temporal reward signals, we employ Double Q-learning with a transformer-based function approximator. Two action-value networks are maintained: an online network $Q(s,a;\phi)$ for action selection and a target network $Q^-(s,a;\psi)$ for evaluation. The target network shares the same architecture but uses separate parameters. The target value is defined as

$$y = r + \gamma Q^- \left( s_{t+1}, \arg\max_a Q(s_{t+1}, a; \phi); \psi \right), \quad (2)$$

where $\gamma$ is the discount factor. The parameters $\phi$ of the online network are optimized by minimizing the squared temporal-difference loss

$$\mathcal{L} = \mathbb{E}\left[ (y - Q(s,a;\phi))^2 \right]. \quad (3)$$

Periodically, the parameters $\phi$ are copied to $\psi$.

### C. Transformer Encoder

Using a standard MLP to estimate $Q(s,a;\phi)$ does not consider the temporal relationships which are strictly necessary for the early detection problem. In contrast, a transformer-based architecture enables direct interactions between all timesteps via self-attention, allowing the model to compare evidence across the entire temporal sequence. This is particularly well-suited for ADR detection, where identifying the onset (or lack) of a persistent signal requires reasoning over long-range dependencies and distinguishing the trend from fluctuations.

### D. Training

The agent enters an episode by sampling a drug–reaction pair. We first construct a temporal sequence of feature vectors, consisting of the states encoding aggregated contingency statistics at all quarters from $t = 1...T$:

$$X = [o_1, o_2, \ldots, o_T]^\top, \quad o_t \in \mathbb{R}^6$$

We then iterate chronologically through each $t = 1...T$, constructing

$$X_t = [o_1, o_2, \ldots, o_t]^\top$$

Each $o_j$ of $X_t$ is projected into a latent space of dimension $d$ via a learned linear embedding, yielding a matrix $Z_t \in \mathbb{R}^{t\times d}$. Positional encodings are added in standard transformer style to preserve temporal order. The sequence is then processed by a series of transformer encoders which produce the hidden representation matrix

$$H_t = \text{Transformer}(Z_t), \qquad H_t \in \mathbb{R}^{t\times d}.$$

We use the final timestep representation $h_t \in \mathbb{R}^d$ as the learned latent belief vector $b_t$, corresponding to the most recent quarter. A small MLP maps this feature vector to action values

$$Q(s_t, \cdot; \phi) = \text{MLP}(h_t), \quad Q(s_t, \cdot; \phi) \in \mathbb{R}^3$$

Training proceeds using Double-Q learning given a transition $(s_t, a_t, r_t, s_{t+1})$ and equations (2), (3). The full training algorithm is concisely summarized in A.

## V. Evaluation

### A. Metrics

We will evaluate the model with a mix of metrics to verify both correctness and timing. Let TP, FP, TN, and FN denote the number of true positives, false positives, true negatives, and false negatives, respectively. Precision, recall, and F1 score are natural to check:

$$\text{Precision} = \frac{\text{TP}}{\text{TP}+\text{FP}}, \qquad \text{Recall} = \frac{\text{TP}}{\text{TP}+\text{FN}},$$

$$\text{F1} = \frac{2 \cdot \text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}$$

Additionally, we must check the timing of detections. For each true positive ADR detection, let $\tau^\star$ denote the reference onset time and $\hat{\tau}$ denote the model detection time. The detection delay is defined as

$$\text{Detection Delay} = \frac{1}{N}\sum_{i=1}^{N} \hat{\tau}_i - \tau_i^\star$$

where $N$ is the number of correctly identified ADR positives. Similarly, the early detection rate (EDR) is defined as

$$\text{EDR} = \frac{1}{N}\sum_{i=1}^{N} \mathbf{1}\left\{\hat{\tau}_i \le \tau_i^\star\right\}.$$

### B. Simulation Setup

We construct the training and evaluation splits by withholding 30% of drug–reaction pairs, along with all corresponding temporal statistics, for validation and testing. The resulting training set contains 672,749 sequences with an average length of 35.8 quarters. Across the full dataset, 7.21% of pairs are labeled as positive ADR associations, highlighting the inherent class imbalance. The held-out test and validation sets consist of 144,019 sequences each. Model selection is performed via validation at each training epoch, with checkpoints chosen based on the highest F1 score subject to an early detection constraint of EDR $\geq 0.70$. We note that our model selection is not a one-off epoch and is stable over many iterations. However, the constraint ensures that improvements in classification performance do not come at the expense of delayed detection, providing a safeguard against the inherent noise in complex Q-learning frameworks applied to large-scale, real data.

### C. Results

TABLE II
PERFORMANCE OF PROPOSED MODEL ON FAERS

| Model | Prec. ↑ | Rec. ↑ | F1 ↑ | Dly. ↓ | EDR ↑ |
|---|---|---|---|---|---|
| Transformer RL | 0.3766 | 0.7008 | 0.4899 | -4.7954 | 0.9382 |

Approximately 70% of labeled positive ADR pairs are successfully identified and 93.8% of true positives are detected prior to the estimated signal emergence time. On average, positive signals are identified nearly five quarters early.

### D. Discussion

The combination of higher recall (0.7008) and relatively lower precision (0.3766) suggests that the model favors early identification over conservative confirmation. This is consistent with the reward structure, which favors early action even under uncertainty. The reward is also designed to prefer higher recall even if precision is mildly scarified, with is in line with pharmacological model expectations.

The negative average delay ($-4.90$ quarters) and strong EDR further reinforce this interpretation, but raise additional questions. To the first point, the results indicate that meaningful signal structure structure exists earlier than statistical methods may detect it, and the transformer hidden state captures this well. However, the balance between early detection and classification performance remains to be investigated further. Namely, the question is whether re-parameterizing the reward to favor a marginally lower EDR could result in meaningfully improved classification performance.

An additional interpretation could be derived from the constructed signal emergency time, $\tau^\star$. The fact that the model consistently predicts before this point suggests that it is learning patterns that precede traditional statistical agreement, but it does not account for the robustness of the label itself. As these methods and their selected thresholds are standard for ADR detection, it suggests that the proposed model is truly uncovering early signal emergence patterns, but repeated testing with $\tau^\star$ constructed from different detection thresholds is necessary to confirm this conclusion.

## VI. CONCLUSION

We propose a transformer-based reinforcement learning approach to detect ADR signals while balancing early detection with accuracy over FAERS. We first propose a method for augmenting reasonable detection time labels in the absence of ground-truth data by considering an ensemble of traditional signal detection methods used on this dataset. By modeling ADR detection as a partially observable Markov decision process and designing a temporal reward function, the agent balances both classification correctness and resource-efficient timing. Empirical results demonstrate strong temporal awareness, correctly flagging over $70\%$ of positives and detecting them early $94\%$ of the time. To this point, we confirm claims that the transformer is a sufficient choice for approximating the belief state in partially observable processes, notably in the case of non-stationary environments and reward functions. These findings highlight the effectiveness of reinforcement learning for time–aware decision over pharmacovigilance data. This is particularly important in FAERS, where machine learning methods are increasingly used to efficiently flag large-scale data for investigation by medical professionals.

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# APPENDIX A
# TRAINING ALGORITHM

**Algorithm 1: Transformer-Based Double Q-Learning**

Initialize $\phi$, $\psi \leftarrow \phi$
**for** each episode **do**
  Sample $(D, R)$ pair
  Construct $X = [o_1, \ldots, o_T]$
  **for** $t = 1$ to $T$ **do**
    $X_t = [o_1, \ldots, o_t]$
    $H_t = \text{Transformer}(X_t)$
    $h_t = H_t[t]$
    $Q(s_t, \cdot; \phi) = \text{MLP}(h_t)$
    Select $a_t$ via $\epsilon$-greedy
    Observe $r_t$, $s_{t+1}$
    $y = r_t + \gamma Q^-(s_{t+1}, \arg\max_a Q(s_{t+1}, a; \phi); \psi)$
    $\mathcal{L} = (y - Q(s_t, a_t; \phi))^2$
    Perform gradient descent on $\phi$
    **if** terminal **then break**
    **end if**
  **end for**
**end for**
Periodically update $\psi \leftarrow \phi$

# APPENDIX B
# ARCHITECTURAL DETAILS & HYPERPARAMETERS

TABLE III
MODEL ARCHITECTURE

| Component | Implementation |
|---|---|
| Backbone | 4 Transformer Encoder layers |
| Model Dimension ($d$) | 128 |
| Attention Heads | 4 |
| FFN Dimension | 512 |
| Q-Head | Linear($d$,$d$) $\rightarrow$ $\phi(x)$ $\rightarrow$ Linear($d$,3) |
| Activation Function | GELU |
| Learning Rate | $5 \times 10^{-5} \rightarrow 1 \times 10^{-5}$ |
| Learning Rate Schedule | Cosine decay |
| Batch Size | 256 |
| Discount Factor ($\gamma$) | 1.0 |
| Gradient Clipping | 1.0 |
| Dropout | 0.0 |
| Weight Decay | $5 \times 10^{-5}$ |
| Drug/Reaction Embedding Dim | 16 |
| Target Q-Net Update Frequency | 2000 steps |
| $\epsilon$-greedy | $1.0 \rightarrow 0.1$ |
| $\epsilon$ Schedule | Exponential decay |
| $\epsilon$ Decay Steps | 250,000 |

TABLE IV
REWARD HYPERPARAMETERS

| Symbol | Description | Value |
|---|---|---|
| $w_{\text{pos}}$ | Positive class weight | 6.5 |
| $w_{\text{neg}}$ | Negative class weight | 0.5 |
| $\rho$ | Wait reward scale | 0.04 |
| $\lambda$ | Wait penalty offset | 0.015 |
| $\sigma$ | TP timing spread | 2.5 |
| $\alpha$ | TP timing shift | 0.9 |
| $\beta_1$ | FP attenuation strength | 0.3 |
| $\beta_2$ | TN scaling factor | 1.0 |
| $\beta_3$ | FN scaling factor | 1.0 |
| $T_1$ | Wait decay timescale | 8.0 |
| $T_2$ | FP decay timescale | 8.0 |
| $T_3$ | TN growth timescale | 6.0 |